Project 2: Convolutional Neural Networks for Colorectal Cancer Histopathology Classification

BME 6938 — Medical AI

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GitHub Repository: https://github.com/wq2581/bme6938-2026-spring-project2

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Abstract

Colorectal cancer (CRC) is the third most commonly diagnosed cancer worldwide, and accurate histopathological classification of tissue types is crucial for diagnosis and treatment planning. In this project, we developed and compared two convolutional neural network (CNN) approaches for automated classification of colorectal tissue into nine histological categories using the PathMNIST dataset from the MedMNIST collection. We implemented a custom 5-block CNN trained from scratch and a pretrained ResNet-18 model fine-tuned on the target task. Both models achieved strong performance, with the Custom CNN reaching 94.96% accuracy (F1=0.949, ROC-AUC=0.996) and ResNet-18 achieving 95.01% accuracy (F1=0.950, ROC-AUC=0.998) on the held-out test set. These results demonstrate that deep learning models can reliably distinguish between diverse colorectal tissue types, with transfer learning providing a marginal advantage in generalization, particularly on underrepresented classes.

1. Introduction

Colorectal cancer is responsible for approximately 900,000 deaths annually, making it the second leading cause of cancer-related mortality worldwide (Sung et al., 2021). Histopathological examination of tissue biopsies remains the gold standard for CRC diagnosis, but manual slide review is time-consuming, subjective, and prone to inter-observer variability (Bera et al., 2019). Computer-aided diagnosis (CAD) systems powered by deep learning offer the potential to assist pathologists by providing rapid, consistent, and objective tissue classification.

The clinical significance of automated histopathology classification extends beyond simple diagnosis. Accurate identification of tissue subtypes — including tumor epithelium, stroma, lymphocytic infiltration, and normal mucosa — provides valuable prognostic information and can guide treatment selection (Kather et al., 2019). For example, the presence of tumor-infiltrating lymphocytes is associated with favorable outcomes in CRC patients, while stromal composition influences treatment response (Galon et al., 2006).

This project addresses the task of nine-class colorectal tissue classification using convolutional neural networks. We compare two modeling approaches: (1) a custom CNN architecture designed from scratch, and (2) a ResNet-18 model pretrained on ImageNet and fine-tuned on pathology images. The intended beneficiary population includes pathologists and clinical laboratories that could integrate such models into their diagnostic workflows to improve throughput and consistency.

Our objectives are to: (a) develop a reproducible deep learning pipeline for histopathology image classification, (b) evaluate the effectiveness of transfer learning versus training from scratch on biomedical image data, and (c) analyze model performance across different tissue types to identify clinical strengths and limitations.

2. Literature Review

Deep learning has transformed computational pathology, with convolutional neural networks achieving expert-level performance on various histopathological classification tasks (Litjens et al.,

2017). The foundational work by Kather et al. (2016) demonstrated that CNNs could classify colorectal cancer tissue textures with high accuracy, establishing a benchmark dataset of 5,000 tissue patches across eight classes. This was later expanded to the NCT-CRC-HE-100K dataset containing 100,000 image patches from hematoxylin and eosin (H&E) stained slides (Kather et al., 2019), which forms the basis of the PathMNIST dataset used in this project.

Transfer learning has proven particularly effective in medical imaging, where labeled data is often scarce relative to natural image datasets. Tajbakhsh et al. (2016) showed that fine-tuning pretrained models consistently outperformed training from scratch on medical image analysis tasks. ResNet architectures (He et al., 2016) have become the de facto standard backbone for histopathology classification due to their ability to train very deep networks through residual connections. Specifically, ResNet-18 and ResNet-50 have been widely adopted in pathology applications (Coudray et al., 2018).

The MedMNIST benchmark (Yang et al., 2023) standardized the evaluation of deep learning models across multiple biomedical image modalities, providing consistent data splits and evaluation protocols. PathMNIST, derived from the NCT-CRC-HE-100K dataset, includes nine tissue classes at various image resolutions, enabling systematic comparison of model architectures.

Data augmentation strategies are critical in medical imaging to mitigate overfitting and improve generalization. Tellez et al. (2019) showed that stain augmentation and geometric transformations significantly improved model robustness in histopathology. Color jitter is especially relevant for H&E-stained images, which exhibit substantial staining variability across laboratories and scanners (Macenko et al., 2009).

Recent work has also explored attention-based architectures such as Vision Transformers (ViT) for pathology (Chen et al., 2022), though CNNs remain competitive, especially when training data is limited. The gap between custom CNNs and pretrained models aligns with the broader literature suggesting that ImageNet pretraining provides useful low-level feature representations that transfer effectively to histopathology (Mormont et al., 2020). Studies have also emphasized the importance of model interpretability and uncertainty estimation in medical AI systems for clinical deployment (Kompa et al., 2021).

3. Methods and Data

3.1 Dataset Description

We used the PathMNIST dataset from the MedMNIST v2 collection (Yang et al., 2023), derived from the NCT-CRC-HE-100K dataset of colorectal cancer histopathology (Kather et al., 2019). The dataset contains 107,180 H&E-stained image patches classified into nine tissue types.

Class	Tissue Type	Train	Val	Test
0	Adipose	10,528	1,180	1,338
1	Background	6,660	740	847
2	Debris	2,692	296	339
3	Lymphocytes	4,980	556	634

4	Mucus	8,148	904	1,035
5	Smooth Muscle	4,700	520	592
6	Normal Colon Mucosa	5,820	648	741
7	Cancer-Associated Stroma	3,332	368	421
8	CRC Adenocarcinoma Epithelium	9,696	1,076	1,233

The standard MedMNIST train/validation/test split was used (89,996 / 10,004 / 7,180 samples). Images were loaded at 28×28 resolution and resized to 224×224 pixels via bilinear interpolation to match pretrained model input requirements.

3.2 Exploratory Data Analysis

A dedicated EDA notebook (notebooks/EDA.ipynb) was created to systematically explore the dataset. Key findings include: (1) Class imbalance — Adipose has ~4× more samples than Debris, and Cancer-Associated Stroma is also underrepresented; (2) Visual diversity — each tissue type exhibits distinct morphological patterns; (3) Color variation — H&E staining intensity varies across patches, motivating color jitter augmentation; (4) Pixel distributions — RGB channels show non-uniform intensity distributions consistent with eosin staining characteristics.

3.3 Data Preprocessing and Augmentation

All images were normalized using ImageNet statistics (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]). Training augmentation included: random horizontal and vertical flips (p=0.5), random rotation (±15°), and color jitter (brightness=0.2, contrast=0.2, saturation=0.2, hue=0.1). Geometric augmentation is justified because tissue orientation is arbitrary in histopathology, and color jitter simulates staining variability. Validation and test sets received only resizing and normalization.

3.4 Model Architectures

Custom CNN (trained from scratch): A 5-block convolutional network with increasing channel depth (32→64→128→256→512). Each block consists of a 3×3 convolution, batch normalization, ReLU activation, and max pooling (2×2). The final block uses adaptive average pooling. The classifier head includes FC(512→256), ReLU, dropout (p=0.5), and output layer (256→9). Total parameters: 1,704,201.

ResNet-18 (pretrained, fine-tuned): A ResNet-18 model initialized with ImageNet-pretrained weights. The original classification head was replaced with FC(512→256)→ReLU→Dropout(0.5)→FC(256→9). All layers were trainable (no frozen backbone). Total parameters: 11,310,153.

3.5 Training Strategy

Both models were trained with Adam optimizer (lr=0.001, weight_decay=1e-4), cross-entropy loss, ReduceLROnPlateau scheduler (factor=0.5, patience=3), and early stopping (patience=5

epochs). Batch size was 32 with a maximum of 20 epochs. Random seeds were fixed (seed=42) across all libraries for reproducibility.

4. Results and Evaluation

4.1 Overall Performance

Metric	Custom CNN	ResNet-18
Accuracy	0.9496	0.9501
Precision (weighted)	0.9505	0.9506
Recall (weighted)	0.9496	0.9501
F1-Score (weighted)	0.9492	0.9496
ROC-AUC (weighted)	0.9960	0.9976

Both models achieved approximately 95% test accuracy. ResNet-18 showed a slight advantage in ROC-AUC (0.998 vs. 0.996), indicating better calibrated probability estimates.

4.2 Per-Class Performance

Class	Precision	Recall	F1-Score
Adipose	0.98	0.98	0.98
Background	1.00	1.00	1.00
Debris	0.78	0.91	0.84
Lymphocytes	0.99	1.00	0.99
Mucus	0.96	0.97	0.97
Smooth Muscle	0.82	0.84	0.83
Normal Colon Mucosa	0.98	0.95	0.97
Cancer-Associated Stroma	0.89	0.72	0.80
CRC Adenocarcinoma Epithelium	0.97	0.98	0.97

Custom CNN per-class results:

ResNet-18 per-class results:

Class	Precision	Recall	F1-Score
Adipose	1.00	0.96	0.98
Background	0.99	1.00	1.00
Debris	0.82	1.00	0.90
Lymphocytes	0.99	0.99	0.99
Mucus	0.95	1.00	0.97
Smooth Muscle	0.88	0.85	0.87
Normal Colon Mucosa	0.95	0.96	0.95
Cancer-Associated Stroma	0.80	0.70	0.75
CRC Adenocarcinoma Epithelium	0.97	0.97	0.97

Both models excelled on Background (F1=1.00), Lymphocytes (F1=0.99), and Adipose (F1=0.98). The most challenging class for both models was Cancer-Associated Stroma (F1=0.75–0.80). ResNet-18 notably improved Debris classification (F1=0.90 vs. 0.84) and Smooth Muscle (F1=0.87 vs. 0.83), suggesting pretrained features better capture subtle texture patterns.

4.3 Training Dynamics

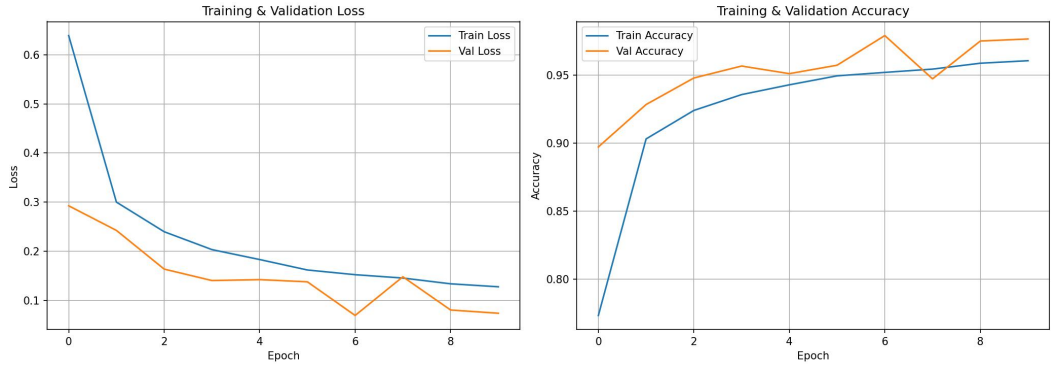

Figure 1. Custom CNN training and validation loss/accuracy over epochs.

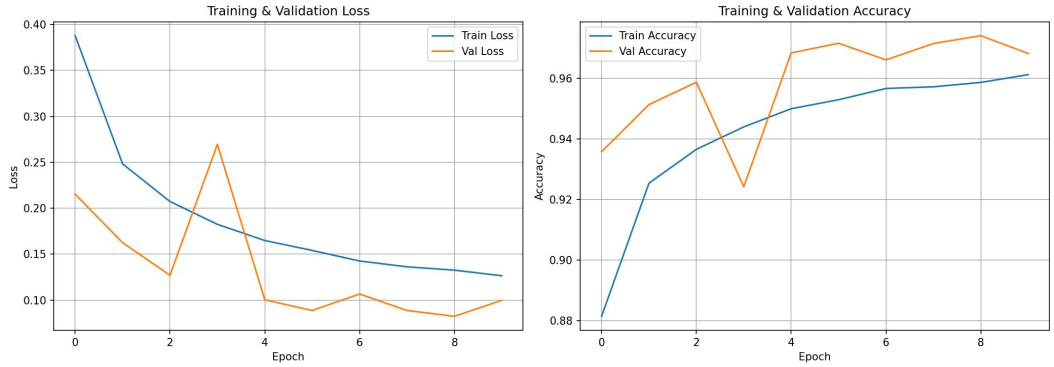

Figure 2. ResNet-18 training and validation loss/accuracy over epochs.

Both models converged within 15–20 epochs. The Custom CNN showed slightly more training loss oscillation, while ResNet-18 exhibited smoother convergence due to well-initialized feature representations from ImageNet pretraining. The learning rate scheduler reduced the learning rate when validation loss plateaued, helping both models escape local minima.

4.4 Confusion Matrix Analysis

Figure 3. Confusion matrix for the Custom CNN on the test set.

Figure 4. Confusion matrix for ResNet-18 on the test set.

The confusion matrices reveal that the primary source of errors is confusion between Cancer-Associated Stroma and Smooth Muscle. This is clinically expected, as both tissue types share fibrous morphological features. Debris was sometimes confused with Mucus, which is also morphologically plausible given their overlapping visual appearance in H&E-stained sections.

4.5 ROC Curve Analysis

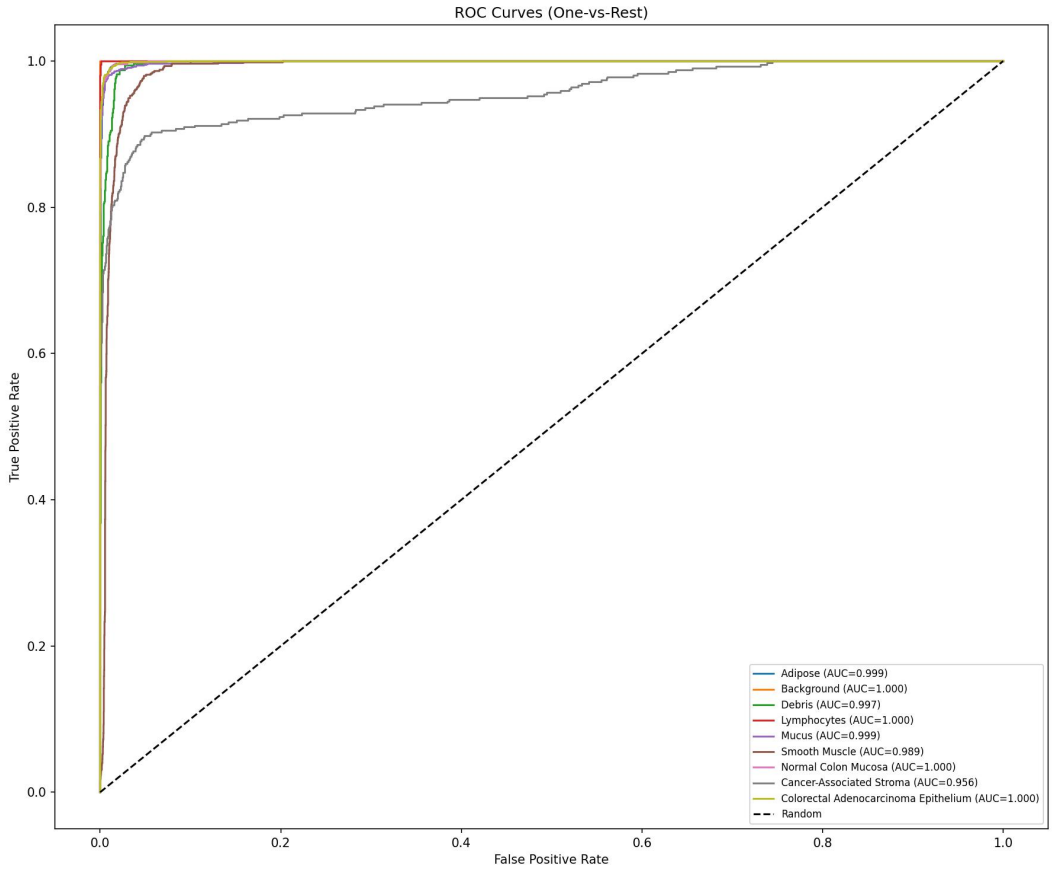

Figure 5. ROC curves (one-vs-rest) for the Custom CNN across all 9 classes.

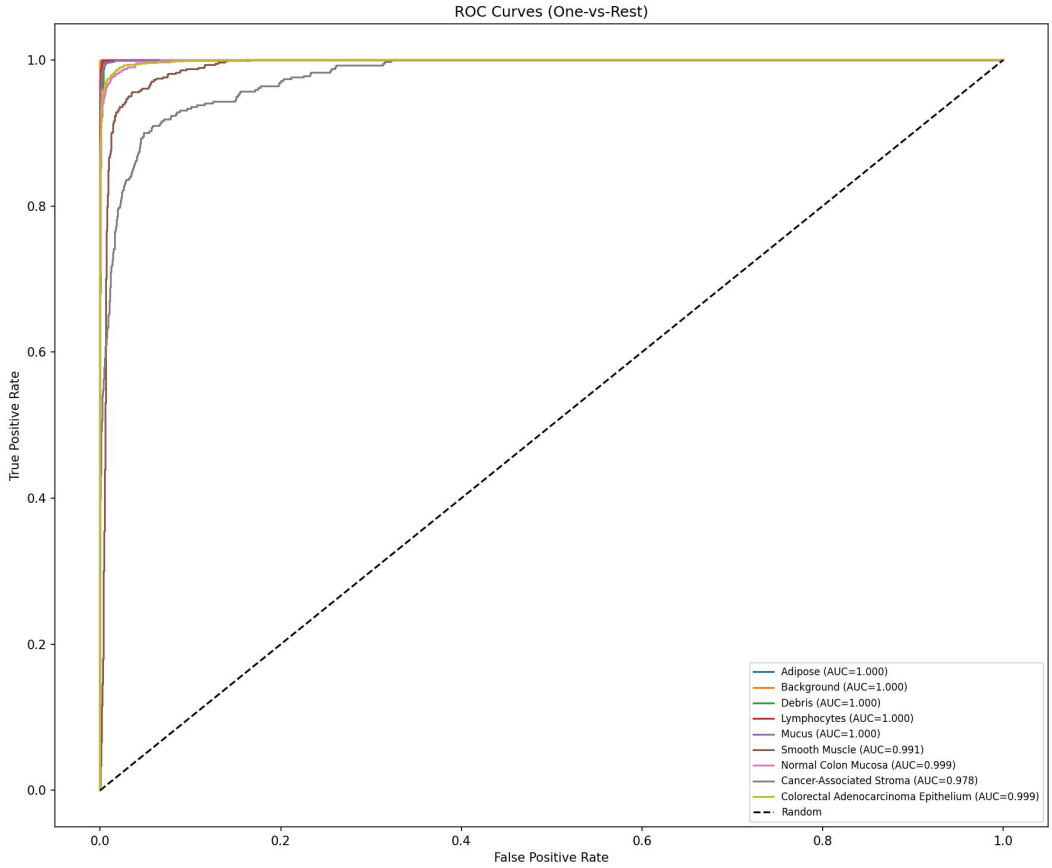

Figure 6. ROC curves (one-vs-rest) for ResNet-18 across all 9 classes.

The ROC curves demonstrate excellent discrimination across all classes, with per-class AUC values exceeding 0.98 for most tissue types. Cancer-Associated Stroma had the lowest individual AUC, consistent with its lower F1-score, though it remained above 0.97 for both models.

5. Discussion and Limitations

5.1 Interpretation of Results

Both models achieved strong classification performance (~95% accuracy), demonstrating that deep learning can effectively distinguish between nine colorectal tissue types. The near-equivalent performance of the Custom CNN and ResNet-18 is noteworthy: while transfer learning is generally expected to provide a significant advantage, the large size of the PathMNIST training set (89,996 samples) allowed the custom CNN to learn competitive representations from scratch.

The marginal ROC-AUC advantage of ResNet-18 (0.998 vs. 0.996) suggests that pretrained features provide better probability calibration, which is clinically important when confidence scores are used for triage or quality control. From a clinical perspective, the high accuracy on Adenocarcinoma Epithelium (F1=0.97) is particularly valuable, as this is the primary tumor class.

5.2 Limitations

Resolution trade-off: Due to memory constraints, images were loaded at 28×28 and resized to 224×224, which may result in loss of fine-grained cellular detail compared to native high-resolution imaging. Single dataset: Results are specific to the NCT-CRC-HE-100K dataset and may not generalize to tissue from different scanners, staining protocols, or patient populations. Class imbalance: The dataset exhibits moderate class imbalance (4:1 ratio). No external validation: The model was evaluated only on the standard MedMNIST test split.

5.3 Ethical Considerations

Deploying AI models in pathology raises important ethical concerns. Bias may arise from the dataset being sourced from a single institution. Patient safety requires that such models serve as decision-support tools rather than autonomous diagnostic systems. Data privacy is addressed by using the publicly available, de-identified MedMNIST dataset. Model confidence thresholds and human-in-the-loop validation are essential safeguards for clinical deployment.

5.4 Future Work

Future directions include: training on native 224×224 resolution images with GPU infrastructure; exploring additional architectures (DenseNet-121, EfficientNet-B0, Vision Transformer); implementing Grad-CAM visualization for model interpretability; addressing class imbalance through focal loss or weighted sampling; validating on external datasets; and incorporating uncertainty estimation to flag low-confidence predictions for pathologist review.

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